

Senior Machine Learning Engineer with 6+ years of expertise in **Deep Learning, Classical ML, NLP, Ads Ranking systems, Recommendation Engines**. Proven success in building **scalable ML architectures**, leading cross-functional teams, and delivering production-grade models. Skilled in driving collaboration, aligning stakeholders and managing timelines to deliver high-impact solutions

SKILLS

Programming languages: Python, Java, C++, PHP, HTML, Javascript, HTML/CSS, $\text{\LaTeX}$, SQL, Presto, Thrift

Development: Git, Docker, Kubernetes, Celery, Apache, uWSGI, AWS Cloud, MongoDB, Neo4j, Redis, Elasticsearch, DevSpace

Softwares/Libraries: Tensorflow, PyTorch, HuggingFace, Jupyter, DVC, SageMaker, OpenCV, Spacy, NLTK, Scikit-Learn, Flask

WORK EXPERIENCE

Meta | SR. MACHINE LEARNING ENGINEER

London, UK | April 2024 – Present

Optimized billion-user-scale **Ads ranking models** by designing novel features, directly boosting **advertiser ROAS**. Drove **Feature engineering** across **Retrieval & Ranking** stages, enhancing ad targeting precision. Built scalable tools, **reducing iteration cycles by 30%**

- **Feature Development: Ads personalization boosting**
 - Built features for User/Advertiser purchase-value affinity. Owned **E2E feature lifecycle** - Ideation to Production
 - Improved **RMSE by 0.5%**, boosted **Model Calibration by 2%**. Drove revenue growth through **increased advertiser adoption**
- **Features Importance(FI) & Correlation Module**
 - Spearheaded experiments evaluating **Permutation** and **Binary Stochastic Filtering** FI algorithm for robustness/consistency
 - Enabled **feature importance for pre-production features**. Facilitates early signal validation, **cut redundant development by 5%**
 - Automated **correlation analysis (Pearson, Spearman, Distance)**, cutting excess features & **model memory usage by 10%**
- **Offline Evaluation Automation: Streamlining Metrics calculation**
 - Boosted developer productivity, **reduced evaluation time by 30% & cut resource usage by 15%** by eliminating redundancies
- **Features Updater Framework: Standardize & automate Feature Engineering processes**
 - Streamlined model candidate generation for **team-level A/B tests**, improving efficiency, reducing variance/errors during updates

VMock | SR. MACHINE LEARNING ENGINEER

Gurgaon, India | July 2018 – March 2024

Led cross-functional teams to **develop and deploy advanced ML models for competency identification and ranking**, across VMock's products. Designed & implemented scalable, reliable **ML delivery system using Docker/Kubernetes(EKS, SageMaker) on AWS Cloud**

- **Resume Bullet Generation with Fine-Tuned LLMs**
 - Fine-tuned **Microsoft's Phi-2 (2.7B)** using **LoRA adapters** to generate resume bullets from skills, industry, and role inputs
 - Optimized training with **Mixed Precision** and **DeepSpeed**, reducing memory usage and training time
- **Skills Recommendation Engine**
 - **Retrieval Stage:** Implemented **3 candidate generators**, **Sentence-BERT** for skills attributes, **Matrix Factorization** to capture co-occurrence, and **Node2Vec + GraphSage** for **Skills Ontology**, procuring **92% Recall** in retrieving **top-300 skills from 40k**
 - **Raking Stage:** Employed **GBDT based LGBMRanker** to predict top-k skills, achieving **87% nDCG@10** on test dataset
- **Skills Detection on resume bullets**
 - Developed a **multi-label classification model** to accurately detect skills from **40K distinct labels** on resume bullets
 - Utilized **Deep Siamese BERT** with **Contrastive Loss**, achieving **80% Recall**. Employed **Bagging** to mitigate class bias/imbalance
- **Skills Rating: Assessing candidate's top skills in resume**
 - Employed **Skills Ontology, Graph Algorithms (Dijkstra & HITS-Rank)**, resume factors to adeptly assess candidate skill relevance

EDUCATION

IIT Delhi

Delhi, India | 2018

BACHELOR OF TECHNOLOGY, ELECTRICAL ENGINEERING

CGPA : 7/10

Coursework: Advanced Machine Learning, Data Structures and Algorithms, Extreme Classification, Reinforcement Learning, Introduction to Machine Learning, Cryptography and Computer Security, **Design Practices in Computer Science**, Probability and Stochastic Process, Operating Systems, Computer Architecture, **Build Basic GANs** (Coursera Certificate)

PROJECTS

Optimizing Electric Power Distribution via Demand-Side Management

GENETIC ALGORITHM, MATLAB

- Devised strategy to reduce power distribution cost & peak load, leveraging day-ahead prices, forecasted load, user preferences
- Employed **Particle Swarm Optimization** for device scheduling, achieving **6% peak load reduction and 15% cost savings**

Detection and Delineation of Word Boundaries in Audio Speech Recordings

BI-LSTM, TENSORFLOW, MFCC

- Built **Bi-LSTM based Supervised classification model**, achieving **70% accuracy** using **MFCC features** from **TIMIT dataset**